

A. Circle of Apple Trees

time limit per test: 1 second
 memory limit per test: 256 megabytes

There are n apple trees arranged in a circle. Each tree bears exactly one apple, and the beauty of the apple on the i -th tree is given by b_i for all $1 \leq i \leq n$. You start in front of tree 1.

At each tree, you may choose to either eat the apple or skip it. After making your choice, you move to the next tree: from tree i , you move to tree $i + 1$ for $1 \leq i \leq n - 1$, and from tree n , you move back to tree 1. This process continues indefinitely as you move through the trees in a cycle.

However, you have a special condition: you may only eat an apple if its beauty is strictly greater than the beauty of the last apple you ate. For example, if $b = [2, 1, 2, 3]$ and you eat the apple on tree 1 (beauty 2), you cannot eat the apples on trees 2 and 3 because their beauties are not greater than 2. However, you may eat the apple on tree 4 since $b_4 = 3 > 2$.

Note that you are allowed to skip an apple when you first encounter it, and you can choose to eat it later on a subsequent cycle.

Your task is to determine the maximum number of apples you can eat if you make optimal decisions on when to eat or skip each apple.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 500$). The description of the test cases follows.

The first line of each test case contains a single integer n ($1 \leq n \leq 100$) — the number of apple trees.

The second line of each test case contains n integers $b_1, b_2, \dots, b_n$ ($1 \leq b_i \leq n$) — the beauty of the apples on the trees.

Note that there are no constraints on the sum of n over all test cases.

Output

For each test case, output a single integer representing the maximum number of apples you can eat.

Example

input	Copy
3 4 2 2 2 2 5 1 4 5 1 2 6 5 4 2 1 2 3	
output	Copy
1 4 5	

Note

In the first test case, since all apples have the same beauty, no matter which apple you eat first, you cannot eat any others afterward. Hence, the maximum number of apples that you can eat is 1.

Codeforces Round 1057 (Div. 2)

Contest is running

01:31:36

Contestant



→ Submit?

Language: Python 3.13.2
 Almost always, if you send a solution on PyPy, it works much faster

Choose file: Choose File No file chosen

Be careful: there is 50 points penalty for submission which fails the pretests or resubmission (except failure on the first test, denial of judgement or similar verdicts). "Passed pretests" submission verdict doesn't guarantee that the solution is absolutely correct and it will pass system tests.

Submit

→ Score table

	Score
Problem A	446
Problem B	669
Problem C	1115
Problem D	1561
Problem E	2007
Problem F	2899
Successful hack	100
Unsuccessful hack	-50
Unsuccessful submission	-50
Resubmission	-50

* If you solve problem on 00:27 from the first attempt

In the second test case, you can eat four apples as follows:

- At tree 1, skip the apple.
- At tree 2, skip the apple.
- At tree 3, skip the apple.
- At tree 4, eat the apple (beauty 1).
- At tree 5, eat the apple (beauty 2).
- At tree 1, skip the apple. You are not allowed to eat the apple as its beauty ($b_1 = 1$) is not greater than the beauty of the last apple you ate ($b_5 = 2$).
- At tree 2, eat the apple (beauty 4).
- At tree 3, eat the apple (beauty 5).

It can be proven that it is not possible to eat all five apples; hence, the maximum number of apples that you can eat is 4.

Supported by



ITMO